Set A. Write C programs for the following problems

1.Write a function isEven, which accepts an integer as parameter and returns 1 if the number is even, and 0 otherwise. Use this function in main to accept n numbers and check if they are even or odd.

Code:

```
#include <stdio.h>
int isEven(int num) {
   if (num % 2 == 0) {
      return 1;
   } else {
      return 0;
   }
}
int main() {
   int n, num;
   printf("Enter how many numbers you want to check: ");
   scanf("%d", &n);
   for (int i = 0; i < n; i++) {
      printf("Enter number %d: ", i + 1);
      scanf("%d", &num);
      if (isEven(num)) {
         printf("%d is Even\n", num);
      } else {
         printf("%d is Odd\n", num);
      }
   }
   return 0;
}
```

Output:

```
Enter how many numbers you want to check: 3
Enter number 1: 2
2 is Even
Enter number 2: 3
3 is Odd
Enter number 3: 9
9 is Odd
```

2.Write a function, which accepts a character and integer n as parameter and displays the next n characters.

Code:

```
#include <stdio.h>
void displayNextChars(char ch, int n) {
   printf("The next %d characters after '%c' are: ", n, ch);
   for (int i = 1; i <= n; i++) {
      printf("%c ", ch + i);
   }
   printf("\n");
}
int main() {
   char ch;
   int n;
   printf("Enter a character: ");
   scanf(" %c", &ch);
   printf("Enter the value of n: ");
   scanf("%d", &n);
   displayNextChars(ch, n);
   return 0;
}
```

Output:

```
Enter a character: A
Enter the value of n: 5
The next 5 characters after 'A' are: B C D E F
```

Set B . Write C programs for the following problems

1. Write a function isPrime, which accepts an integer as parameter and returns 1 if the number is prime and 0 otherwise. Use this function in main to display the first 10 prime numbers.

Code:

```
#include <stdio.h>
int isPrime(int num) {
   if (num <= 1) return 0;
   for (int i = 2; i * i <= num; i++) {
      if (num % i == 0) {
         return 0;
      }
   }
   return 1;
}
int main() {
   int count = 0;
   int num = 2;
   printf("The first 10 prime numbers are:\n");
   while (count < 10) {
      if (isPrime(num)) {
         printf("%d ", num);
         count++;
      }
      num++;
   }
   printf("\n");
   return 0;
}
```

Output:

```
The first 10 prime numbers are:
2 3 5 7 11 13 17 19 23 29
```

2. Write a function that accepts a character as parameter and returns 1 if it is an alphabet, 2 if it is a digit and 3 is it is a special symbol. In main, accept characters till the user enters EOF and use the function to count the total number of alphabets, digits and special symbols entered.

Code:

```
#include <stdio.h>
int checkCharacterType(char ch) {
   if ((ch >= 'a' && ch <= 'z') || (ch >= 'A' && ch <= 'Z')) {
```

```c
        return 1;
    } else if (ch >= '0' && ch <= '9') {
        return 2;
    } else {
        return 3;
    }
}
int main() {
    char ch;
    int alphabetCount = 0, digitCount = 0, specialCount = 0;
    printf("Enter characters (Press Ctrl+D or Ctrl+Z to stop):\n");
    while ((ch = getchar()) != EOF) {
        if (ch == '\n' || ch == '\r') {
            continue;
        }
        int type = checkCharacterType(ch);
        if (type == 1) {
            alphabetCount++;
        } else if (type == 2) {
            digitCount++;
        } else if (type == 3) {
            specialCount++;
        }
    }
    printf("\n--- Final Counts ---\n");
    printf("Total Alphabets: %d\n", alphabetCount);
    printf("Total Digits: %d\n", digitCount);
    printf("Total Special Symbols: %d\n", specialCount);
    return 0;
}
```

Output:

```
Enter characters (Press Ctrl+D or Ctrl+Z to stop):
Hello 123!@
--- Final Counts ---
Total Alphabets: 5
Total Digits: 3
Total Special Symbols: 2
```

3. **Taylor Series Calculation ($\sin(x)$)** Use custom power and factorial functions to calculate the sum of the first n terms of the Taylor series for $\sin(x)$.

- **Mathematical Formula Used:**

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

Code:

```
#include <stdio.h>
double power(double base, int exp) {
   double result = 1.0;
   for (int i = 0; i < exp; i++) {
      result *= base;
   }
   return result;
}
double factorial(int n) {
   double fact = 1.0;
   for (int i = 1; i <= n; i++) {
      fact *= i;
   }
   return fact;
}
int main() {
   double x, degree, sum = 0.0;
   int n, sign = 1;
   printf("Enter the angle in degrees: ");
   scanf("%l_f", &degree);
   printf("Enter the number of terms (n): ");
   scanf("%d", &n);
   x = degree * (3.1415926535 / 180.0);
   for (int i = 0; i < n; i++) {
      int currentTermPower = 2 * i + 1;
      double term = power(x, currentTermPower) / factorial(currentTermPower);
      sum += sign * term;
      sign = -sign;
   }
   printf("The calculated value of sin(%.2f) using %d terms is: %f\n", degree, n, sum);
```

```
    return 0;
}
```

Output:

```
Enter the angle in degrees: 30
Enter the number of terms (n): 3
The calculated value of sin(30.00) using 3 terms is: 0.500002
```